

DATASCI 406 FA 26, Quiz 2.1

Name	Uniqname	UMID
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Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

For a random variable X and a probability p , what does the quantile function $Q_X(p)$ return?

1. The probability that X is less than or equal to p .
2. The smallest value x such that the CDF evaluated at x is at least p .
3. The value x at which the density of X equals p .
4. The value of the CDF evaluated at X .

Question 2 (1 point)

X has CDF $F(x) = x^3$ for $0 \leq x \leq 1$. What is its quantile function $Q(u)$ for $0 \leq u \leq 1$?

1. u^3
2. $3u^2$
3. $1 - u^{1/3}$
4. $u^{1/3}$

Question 3 (1 point)

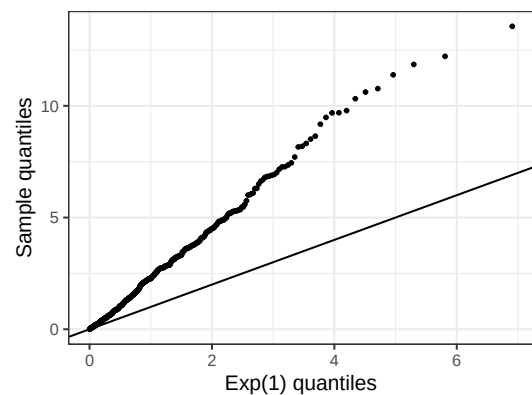
Which line of R produces 1000 draws from an exponential distribution with rate 2, using `qexp()` and not `rexp()`?

1. `runif(1000, qexp(0, rate = 2), qexp(1, rate = 2))`
2. `qexp(1000, rate = 2) * runif(1000)`
3. `qexp(runif(1000), rate = 2)`
4. `runif(qexp(1000, rate = 2))`

Question 4 (1 point)

A sampler is meant to draw from $\text{Exp}(1)$. The QQ plot at right compares 500 of its draws with the $\text{Exp}(1)$ quantiles, and the line is the 45 degree line. Is the sampler correct?

1. Correct: the points form a straight line.
2. Wrong: the points do not follow the 45 degree line.
3. Can't tell: a plot is not a hypothesis test.



Question 5 (1 point)

X takes the values below. To draw X you generate $u = 0.53$ and apply the inverse transform method for discrete variables. What value x do you generate?

x	10	20	30	40
$\Pr(X = x)$	0.2	0.3	0.1	0.4

1. 10
2. 20
3. 30
4. 40